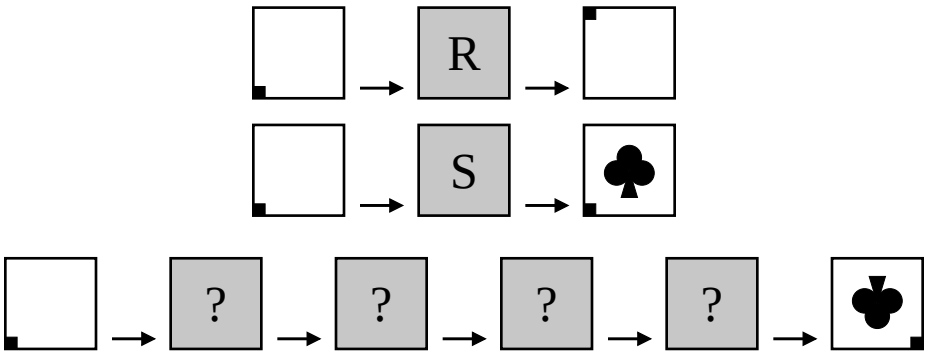
Math Kangaroo Practice

Grade 3-4 · 21 questions · harder · seed 65702

Part A: Each question is worth 3 points

1. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Tom uses the machines 4 times in a row and gets the picture at the end of the bottom row. In which order does Tom use the machines?

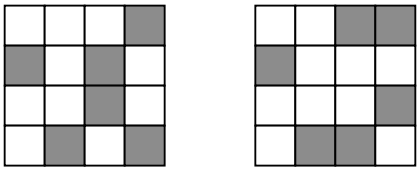


A. SSRR ***B.*** SRRS ***C.*** SRRR ***D.*** RSRR ***E.*** SSSS

2. Four children, Nora, Max, Clemens and Ben, run a race. There are no ties. Nora is not the first. Max and Ben are not next to each other. Nora is not the third. In which order can they be?

A. Clemens, Max, Ben, Nora ***B.*** Clemens, Nora, Ben, Max ***C.*** Ben, Clemens, Nora, Max
D. Max, Nora, Ben, Clemens ***E.*** Clemens, Ben, Max, Nora

3. Tom coloured the left grid. A friend guessed the right grid. How many cells did the friend guess correctly?

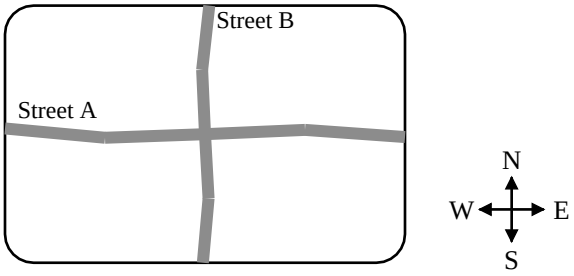


A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

4. Four children, Tom, Nora, Paul and Emma, enter a room one after the other. Nora is not the fourth. Emma is not the first. Tom is not the first. Paul is not the second. How many different orders are there for the children to enter the room?

A. 8 ***B.*** 9 ***C.*** 10 ***D.*** 20 ***E.*** 24

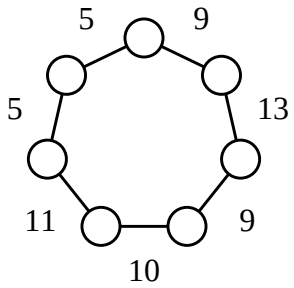
5. North of street A there are 6 houses. East of street B there are 7 houses. South of street A there are 8 houses. No house stands on a street. How many houses are there in the village altogether?



A. 7 houses ***B.*** 12 houses ***C.*** 13 houses ***D.*** 14 houses ***E.*** 15 houses

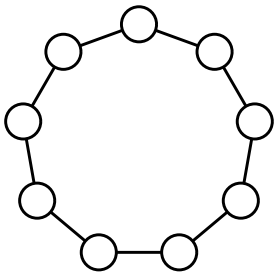
Part B: Each question is worth 4 points

6. Each number between two circles is the sum of the numbers in these two circles. What do the numbers in all the circles add up to?



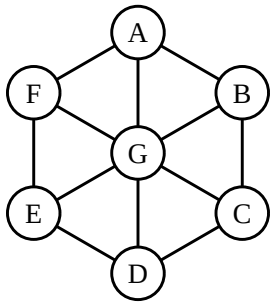
A. 25 ***B.*** 26 ***C.*** 29 ***D.*** 30 ***E.*** 31

7. 9 children stand in a ring and hold hands, as in the picture. Two children who hold hands must wear T-shirts of different colours. What is the smallest number of colours needed?



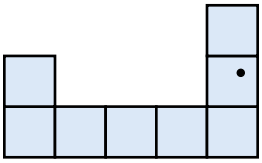
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 9

8. A zoo has 7 animals, A to G. A line between two animals means that they do not get along, so they must not live in the same cage. Any number of animals can live in one cage. What is the smallest number of cages the zoo needs?



A. 2 cages ***B.*** 3 cages ***C.*** 4 cages ***D.*** 7 cages ***E.*** 12 cages

9. Using the pieces A, B, C, D and E you can fill this shape completely. The pieces may be turned. Which of the pieces lies on the dot?

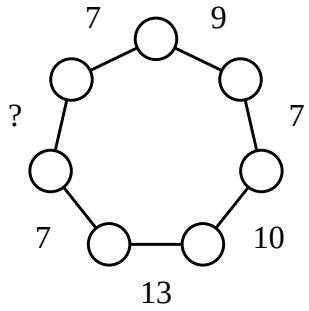


A. ***B.*** ***C.*** ***D.*** ***E.***

10. Maria sets a secret code with 5 digits on a bike lock. Five friends guess the code: 14326, 74346, 74926, 74326 and 74320. Maria says: "One of you is right. Each of the others has exactly 4 digits right, in the right place." What is the code?

A. 14326 ***B.*** 74320 ***C.*** 74346 ***D.*** 74326 ***E.*** 74926

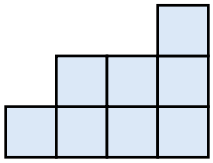
11. The numbers from 1 to 7 are written into the circles, each exactly once. Each number between two circles is the sum of the numbers in these two circles. Which number belongs where the question mark is?



A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 7 ***E.*** 56

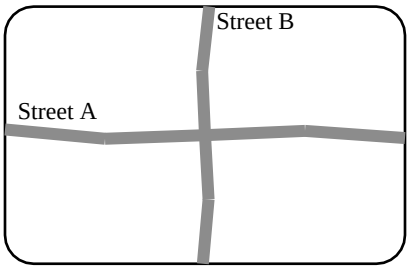
Part C: Each question is worth 5 points

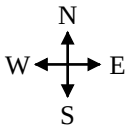
12. Four of the pieces A, B, C, D and E fill this shape completely. The pieces may be turned. Which piece is left over?



A. ***B.*** ***C.*** ***D.*** ***E.***

13. North of street A there are 12 houses. South of street A there are 12 houses. East of street B there are 11 houses. 8 of the houses are north of street A **and** east of street B. How many houses are south of street A **and** west of street B?





A. 9 ***B.*** 10 ***C.*** 11 ***D.*** 12 ***E.*** 13

14. Four children, Nora, Ben, Leo and Hermann, run a race. There are no ties. Leo is not the third. Ben is not the second. Hermann is somewhere before Leo. Which extra sentence makes sure that there is exactly **one** possible order?

A. Hermann is somewhere before Ben.

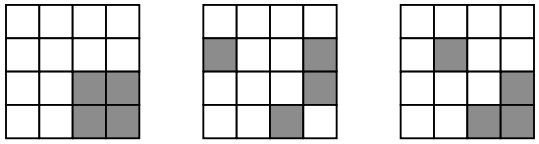
B. Hermann comes right after Nora.

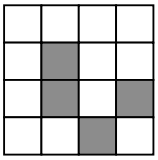
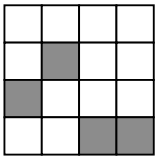
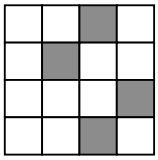
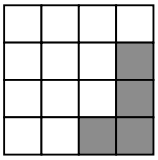
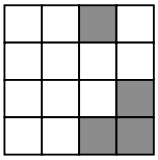
C. Hermann is the fourth.

D. Nora is the fourth.

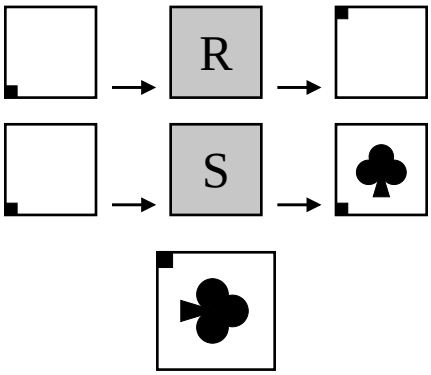
E. Hermann is not the second.

15. Maria colours exactly 4 cells of a 4 × 4 grid grey. Three friends guess (top row). Maria says: "Nobody is right, but each of you guessed exactly 3 cells correctly." Which of the patterns A to E did Maria colour?



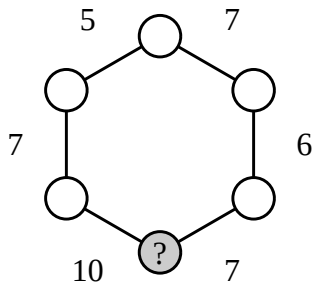
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

16. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Leo uses machine S once and machine R as often as needed. How many times at least must Leo use machine R to get this picture?



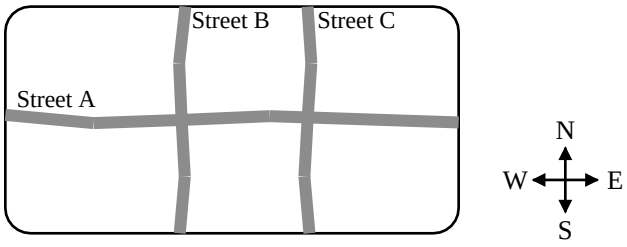
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

17. The numbers from 1 to 6 are written into the circles, each exactly once. Each number between two circles is the sum of the numbers in these two circles. Which number is in the circle with the question mark?



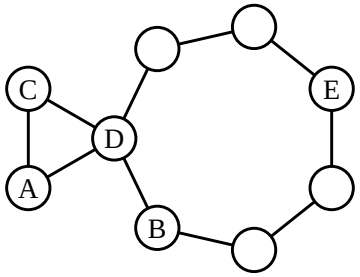
A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 10

18. Streets B and C both go from north to south (see map). North of street A there are 16 houses. South of street A there are 8 houses. West of street B there are 12 houses. East of street C there are 6 houses. How many houses are there between street B and street C?



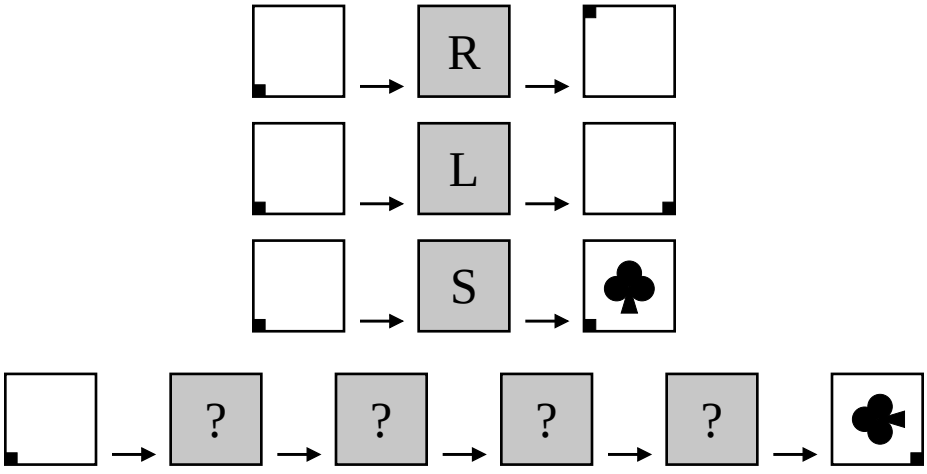
A. 2 houses ***B.*** 5 houses ***C.*** 6 houses ***D.*** 8 houses ***E.*** 12 houses

19. Anna has only **two** colours. Circles joined by a line must get different colours. Anna may take away one circle together with its lines. Which circle must Anna take away, so that two colours are enough for the other circles?



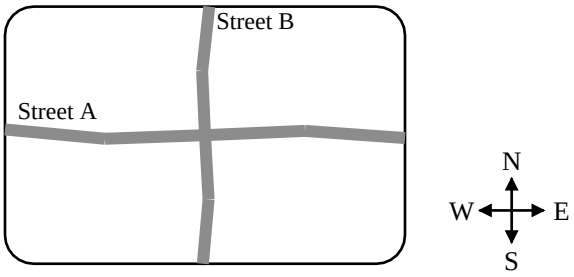
A. circle A ***B.*** circle B ***C.*** circle C ***D.*** circle D ***E.*** circle E

20. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Machine L turns the sheet a quarter turn anticlockwise. Leo uses the machines 4 times in a row and gets the picture at the end of the bottom row. Which order works?



A. RRSL ***B.*** LRSS ***C.*** SLLR ***D.*** RSRR ***E.*** SLSS

21. North of street A there are 13 houses. South of street A there are 13 houses. East of street B there are 11 houses. 6 of the houses are north of street A **and** east of street B. How many houses are south of street A **and** west of street B?



A. 8 ***B.*** 9 ***C.*** 10 ***D.*** 13 ***E.*** 15

Answer Key

1.	**D**	**2.**	**D**	**3.**	**A**	**4.**	**A**	**5.**	**D**	**6.**	**E**
7.	**C**	**8.**	**B**	**9.**	**C**	**10.**	**D**	**11.**	**B**	**12.**	**D**
13.	**A**	**14.**	**D**	**15.**	**D**	**16.**	**B**	**17.**	**A**	**18.**	**C**
19.	**D**	**20.**	**C**	**21.**	**A**						

Solutions

1. D (RSRR) — The corner mark turned 3 quarter turns, so machine R is used 3 times. The club turned 2 quarter turns, so 2 R come after S: RSRR. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · which_order]

2. D (Max, Nora, Ben, Clemens) — Check each order against every rule. Only Max, Nora, Ben, Clemens keeps all of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · which_order]

3. A (3) — Look for cells that are grey in both grids: 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · how_many_right]

4. A (8) — Write down all 24 orders, for example sorted by who is first. Then cross out every order that breaks a rule. 8 orders are left. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · not_rules]

5. D (14 houses) — Every house is either north or south of street A, so the village has 6 + 8 = 14 houses. The east houses are already counted in these. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · whole_village]

6. E (31) — Add all the numbers between the circles: 9 + 13 + 9 + 10 + 11 + 5 + 5 = 62. Every circle is in two of these sums, so all the circles together make 62 ÷ 2 = 31. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · all_together]

7. C (3) — Red, blue, red, blue, ... works until the last child: with 9 children (an odd number) the last child holds hands with two different colours. So 3 colours are needed. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · hand_in_hand]

8. B (3 cages) — A cage is like a colour: animals joined by a line need different cages. Look for animals that are all joined to each other. Then share the animals out step by step: 3 cages are enough, and fewer do not work. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · zoo_cages]

9. C — Start where there is the least room: the corners and the narrow parts of the shape. Slanted edges must meet slanted edges. When all pieces are in, piece C lies on the dot. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · dot_piece]

10. D (74326) — Compare each guess with all the others, digit by digit. Only 74326 has exactly 4 digits in the same places as every other guess. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · lock_code]

11. B (3) — 1 + 2 + ... + 7 = 28. Every circle is in two sums, so all the sums together make 2 × 28 = 56. The shown sums make 53, so the missing one is 56 – 53 = 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · hidden_sum]

12. D — Pieces B and D have the same number of half squares, so counting alone does not decide. Fit the pieces in, starting in the corners: piece D is not needed. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · left_over]

13. A (9) — The village has 12 + 12 = 24 houses, so 24 – 11 = 13 are west of street B. North-west there are 12 – 8 = 4 houses. So south-west there are 13 – 4 = 9. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · one_corner]

14. D (Nora is the fourth.) — First list the 6 orders that keep the rules: Nora, Hermann, Ben, Leo; Ben, Nora, Hermann, Leo; Ben, Hermann, Nora, Leo; Hermann, Nora, Ben, Leo; Hermann, Leo, Nora, Ben; Hermann, Leo, Ben, Nora. Only the sentence "Nora is the fourth." leaves exactly one of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · one_more_rule]

15. D — Check each pattern against all three guesses: pattern D has exactly 3 cells in common with each of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · not_a_guess]

16. B (1) — The club is turned 1 quarter turn, so 1 R comes after S. The mark needs 0 more quarter turns before S. Together 1. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · fewest_turns]

17. A (6) — With 6 circles every guess closes the ring, so check something else: the numbers must be 1 to 6, each once. Only 6 gives 6, 4, 3, 2, 5, 1. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · even_ring]

18. C (6 houses) — The village has 16 + 8 = 24 houses. Take away the houses west of B and east of C: 24 – 12 – 6 = 6 houses are left between the two streets. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · three_streets]

19. D (circle D) — A ring with an odd number of circles cannot be coloured with two colours taking turns. There are two such rings (3 and 7 circles). Only circle D belongs to both rings, so taking it away breaks both of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · take_one_away]

20. C (SLLR) — Check each order: follow the corner mark through every machine and turn the club only with the machines after S. Only SLLR gives the picture. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · left_machine]

21. A (8) — The village has 13 + 13 = 26 houses, so 26 – 11 = 15 are west of street B. North-west there are 13 – 6 = 7 houses. So south-west there are 15 – 7 = 8. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · one_corner]